

Lecture 9 (153-248)

Recap (last 2 lectures)

Data: $y_0, y_1, \dots, y_{n-1}$

$$y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t + \varepsilon_t$$

$\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

$f \rightarrow$ Frequency $\in [0, \frac{1}{2}]$

$\beta_0, \beta_1, \beta_2, \sigma$: Nuisance parameters

⑦ $RSS(f)$

$$= \min_{\beta_0, \beta_1, \beta_2} \sum_{t=0}^{n-1} (y_t - \beta_0 - \beta_1 \cos 2\pi f t - \beta_2 \sin 2\pi f t)^2$$

$$= \|y - X_f \hat{\beta}_f\|^2$$

$$y = \begin{pmatrix} y_0 \\ \vdots \\ y_{n-1} \end{pmatrix} \quad X_f = \begin{pmatrix} 1 & \cos 2\pi f(0) & \sin 2\pi f(0) \\ \vdots & \cos 2\pi f(1) & \sin 2\pi f(1) \\ \vdots & \vdots & \vdots \\ 1 & \cos 2\pi f(n-1) & \sin 2\pi f(n-1) \end{pmatrix}$$

$n \times 3$

$$\hat{\beta}_f = (X_f^T X_f)^{-1} X_f^T y$$

To estimate f : minimize $RSS(f)$

uncertainty
quantification:

$$\text{post}(f) \propto \left[\frac{1}{\text{RSS}(f)} \right]^{\frac{n-3}{2}} |X_f^T X_f|^{-1/2} \mathbb{I}(0 < f < \frac{1}{2})$$

Two ways of discretization:

- ① Take some dense grid of values of f .
- ② Grid of Fourier frequencies
 $(\frac{1}{n}, \frac{2}{n}, \dots)$ between 0 & $\frac{1}{2}$

Advantage of Fourier in computation.

① Compute DFT of $y_0 \dots y_{n-1}$
 $n \cdot \text{fft} \cdot \text{fft}^*(y)$ in $O(n \log n)$ time

can be complex
 $\rightarrow b_j = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t)$

$j=0, 1, \dots, n-1$

$b_{n-j} = \overline{b_j}$

② Periodogram

$$I(\frac{j}{n}) = \frac{|b_j|^2}{n} \quad \text{for } 0 < \frac{j}{n} < \frac{1}{2}$$

③ RSS:

$$\text{RSS}(\frac{j}{n}) = \sum_{t=0}^{n-1} (y_t - \bar{y})^2 - 2 I(\frac{j}{n})$$

for $0 < \frac{j}{n} < \frac{1}{2}$

$$\downarrow \quad X_f^T X_f = \begin{bmatrix} n & 0 & 0 \\ 0 & n/2 & 0 \\ 0 & 0 & n/2 \end{bmatrix}$$

④ Minimize $RSS(\frac{j}{n})$ over j to estimate $\hat{f} = \frac{j}{n}$

$$\textcircled{5} \text{ post}(\frac{j}{n}) \propto \left(\frac{1}{RSS(j/n)} \right)^{\frac{n-3}{2}} |X_{\frac{j}{n}}^T X_{\frac{j}{n}}|^{-1/2}$$

$$\propto \left(\frac{1}{RSS(j/n)} \right)^{\frac{n-3}{2}}$$

Multiple Frequency Models

$$y_t = \beta_0 + \beta_1 \cos 2\pi f_1 t + \beta_2 \sin 2\pi f_1 t + \beta_3 \cos 2\pi f_2 t + \beta_4 \sin 2\pi f_2 t + \varepsilon_t$$

$\varepsilon_t \stackrel{i.i.d}{\sim} N(0, \sigma^2)$

$$\boxed{f_1, f_2}$$

$$RSS(f_1, f_2) = \|y - X_{f_1, f_2} \hat{\beta}_{f_1, f_2}\|^2$$

$$X_{f_1, f_2} = \begin{pmatrix} 1 & \cos 2\pi f_1 t & \sin 2\pi f_1 t & \cos 2\pi f_2 t & \sin 2\pi f_2 t \\ \vdots & & & & \\ 1 & & & & \end{pmatrix}$$

$\downarrow$
 $n \times 5$

$$\text{Post}(f_1, f_2) \propto \left(\frac{1}{\text{RSS}(f_1, f_2)} \right)^{\frac{n-5}{2}} \left| X_{f_1, f_2}^T X_{f_1, f_2} \right|^{-1/2}$$

- ① Work with a dense grid of (f_1, f_2)
- ② Restrict to Fourier Frequencies.

Computation for Fourier Frequencies

$$\text{RSS}(f) = \sum (y_t - \bar{y})^2 - 2I(f)$$

→ single sinusoid
f Fourier $\in (0, \frac{1}{2})$

$$\text{RSS}(f_1, f_2) = \sum (y_t - \bar{y})^2 - 2I(f_1) - 2I(f_2)$$

→ f_1, f_2 distinct
Fourier frequencies
in $(0, \frac{1}{2})$

ORTHOGONALITY OF SINUSOIDS AT FOURIER FREQUENCIES

$$\begin{aligned} \underset{\substack{\uparrow \\ \text{complex} \\ \text{exponential} \\ \text{sinusoid at} \\ \text{frequency } \frac{j}{n}}}{u^j} &= \left(\exp(2\pi i \frac{j}{n} t) : t=0, 1, \dots, n-1 \right) \\ &= \left(1, \exp(2\pi i \frac{j}{n} 1), \exp(2\pi i \frac{j}{n} 2), \dots, \exp(2\pi i \frac{j}{n} (n-1)) \right) \end{aligned}$$

$j = 0, 1, \dots, n-1$

$$u^{n-j} = \exp\left(2\pi i \frac{(n-j)t}{n}\right) = \underbrace{\exp(2\pi i t)}_1 \underbrace{\exp\left(-2\pi i \frac{j}{n}t\right)}$$

$$u^{n-j} = \overline{u^j}$$

ORTHOGONALITY FACT: $u^0, u^1, \dots, u^{n-1}$ are orthogonal in $\mathbb{C}^n$.

$$\textcircled{\star} \quad \langle u^j, u^k \rangle = \begin{cases} 0 & \text{if } j \neq k \\ n & \text{if } j = k \end{cases}$$

a, b complex vectors

$$\langle a, b \rangle = \sum_{j=1}^n a_j \overline{b_j} \quad \langle a, a \rangle$$

Proof of $\textcircled{\star}$

$$\begin{aligned} & \langle u^j, u^k \rangle \\ &= \sum_{t=0}^{n-1} \exp\left(2\pi i \frac{j}{n}t\right) \overline{\exp\left(2\pi i \frac{k}{n}t\right)} \end{aligned}$$

$$= \sum_{t=0}^{n-1} \exp\left(\frac{2\pi i(j-k)}{n} t\right)$$

$$\begin{aligned} & \downarrow r = \exp\left(\frac{2\pi i(j-k)}{n}\right) \rightarrow \\ & = \sum_{t=0}^{n-1} r^t = \frac{r^n - 1}{r - 1} \end{aligned}$$

$$= \frac{\exp(2\pi i(j-k)) - 1}{r - 1}$$

$$= \frac{\cos 2\pi(j-k) + i \sin 2\pi(j-k) - 1}{r - 1}$$

$$= 0$$

Orthogonality for cosines & sines

$$c^j = \left(\cos\left(2\pi \frac{j}{n} t\right), t=0, 1, \dots, n-1 \right)$$

$$s^j = \left(\sin\left(2\pi \frac{j}{n} t\right), t=0, 1, \dots, n-1 \right)$$

$$0 < \frac{j}{n} < \frac{1}{2}$$

$$\text{or } 0 < j < \frac{n}{2}$$

c^0

n odd

$c^0, c^1, s^1, c^2, s^2, \dots, c^{\frac{n-1}{2}}, s^{\frac{n-1}{2}}$

→ Orthogonal as vectors in n-dimensions

$j \neq k$

$$\langle c^j, c^k \rangle$$

$$= \sum \cos\left(2\pi \frac{j}{n} t\right) \cos\left(2\pi \frac{k}{n} t\right)$$

n even

$c^0, c^1, s^1, \dots, c^{\frac{n-2}{2}}, s^{\frac{n-2}{2}}, c^{\frac{n}{2}}$

→ Orthogonal as vectors in n-dimensions

$$\text{RSS}(\underbrace{f_1, f_2}_{\text{Fourier}}) = \sum (y_t - \bar{y})^2 - 2I(f_1) - 2I(f_2)$$

$$X_{f_1, f_2} = \begin{bmatrix} c^0 & c^j & s^j & c^k & s^k \end{bmatrix}$$

$$f_1 = \frac{j}{n}$$

$$f_2 = \frac{k}{n}$$

$$X_{f_1, f_2}^T X_{f_1, f_2} = \begin{bmatrix} \langle c^0, c^0 \rangle & \langle c^0, c^j \rangle & \langle c^0, s^j \rangle & \langle c^0, c^k \rangle & \langle c^0, s^k \rangle \\ \langle c^j, c^0 \rangle & \langle c^j, c^j \rangle & \langle c^j, s^j \rangle & \langle c^j, c^k \rangle & \langle c^j, s^k \rangle \\ \langle s^j, c^0 \rangle & \langle s^j, c^j \rangle & \langle s^j, s^j \rangle & \langle s^j, c^k \rangle & \langle s^j, s^k \rangle \\ \langle c^k, c^0 \rangle & \langle c^k, c^j \rangle & \langle c^k, s^j \rangle & \langle c^k, c^k \rangle & \langle c^k, s^k \rangle \\ \langle s^k, c^0 \rangle & \langle s^k, c^j \rangle & \langle s^k, s^j \rangle & \langle s^k, c^k \rangle & \langle s^k, s^k \rangle \end{bmatrix}$$

$$= \begin{bmatrix} n & n/2 & 0 \\ 0 & n/2 & n/2 \\ 0 & 0 & n/2 \end{bmatrix} \quad f_1, f_2 \in (0, \frac{1}{2})$$

$$RSS(f_1, f_2) = \text{variance} - 2I(\frac{j}{n}) - 2I(\frac{k}{n})$$

Change of Slope (Broken Stick) Regression

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-c) + \varepsilon_t$$

$RSS(c) \quad c \in \{1, \dots, n\}$

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-c_1) + \beta_3 \text{ReLU}(t-c_2) + \beta_4 \text{ReLU}(t-c_3) + \dots + \beta_{k+1} \text{ReLU}(t-c_k) + \varepsilon_t$$

$$\boxed{c_1 \dots c_k}$$

$$\downarrow$$

$$k=10$$

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-1) + \beta_3 \text{ReLU}(t-2) + \beta_4 \text{ReLU}(t-3) + \dots + \beta_n \text{ReLU}(t-(n-1)) + \varepsilon_t$$

